

LAB 03

PH142 Summer 2026

Announcements

- 7/13: Quiz06, Lab03
- 7/14: Quiz07
- 7/15: Quiz08
- 7/16: Midterm 1
- 7/17: Quiz09, Data Project Part 1



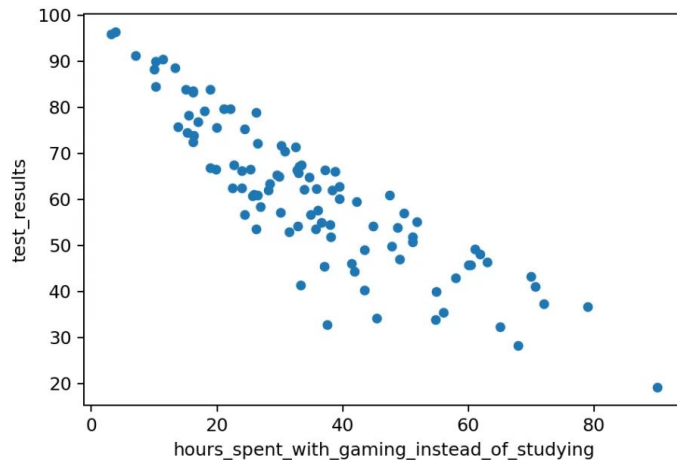
Lab03 Lecture Review

Scatterplots

1. Direction – Is it positive or negative?
2. Form – Is it linear or nonlinear?
3. Strength – Is it strong or weak?
4. Outliers – Are there any obvious ones?

How would you interpret this scatterplot?

Can you think of any confounding factors?



Lab03 Lecture Review

Scatterplots

`geom_point` is used for scatter plots:

Ex. `name of plot <- ggplot(data = dataset, aes(x=var, y=var)) +
 geom_point(na.rm = TRUE) +
 theme_minimal(base_size = 15) +
 labs (x= "", y= "", title= "")`

- To color the points by a variable include `col=variable`
- To create separate plots for combinations of levels of 2 vars i.e. (gender) use `facet_wrap` ex. `facet_wrap ( ~ gender)`

Lab03 Lecture Review

Correlation

- Correlation measures the **strength** and **direction** of a linear relationship between two quantitative variables
- Also written as **r**
- Takes values between -1 to 1 inclusively

Lab03 Lecture Review

Intro to Linear Regression

- **Regression:** Straight line fitted to data to minimize distance between the data and fitted line
 - “Line of best fit” = $a + bx$
 - a = **intercept** (Predictive Value of y when $x=0$)
 - b = **slope** $r^*(s_y/s_x)$
- Interpretation
 - Intercept : the value of the outcome when $x = 0$
 - Slope: For a one-unit change in x , the outcome changes by [number] [units]

Lab03 Lecture Review

Intro to Linear Regression

- `lm()` is the function for a linear model
 - `lm(formula = y ~ x, data = your_dataset)`
 - Add regression line to a scatterplot using `geom_abline()`
 - `glance(data_lm)` will give r squared
- **Interpretation of `lm()`:** A one unit change in **X** is associated with a ____ increase/decrease of **Y**.
- **Interpretation of r-squared:** the fraction of the variation in the values of y that is explained by the line of best fit (the regression of y on x)
- Find **correlation** in data using `summarize(corr_variable = cor(var1, var2))`



LAB 03 Walkthrough

Lab Submission

- Follow the directions on the LAB03 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
 - Copy and paste the given line into the terminal
 - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**